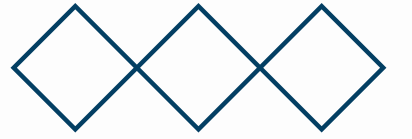


AUTONOMOUS FRAMEWORK FOR REAL- TIME DATA QUALITY, PRIVACY, AND SCHEMA DRIFT MANAGEMENT IN STREAMING ENVIRONMENTS

PRESENTED BY 25-26J-090



OUR TEAM



Amodhya



Lavindi



Amanda

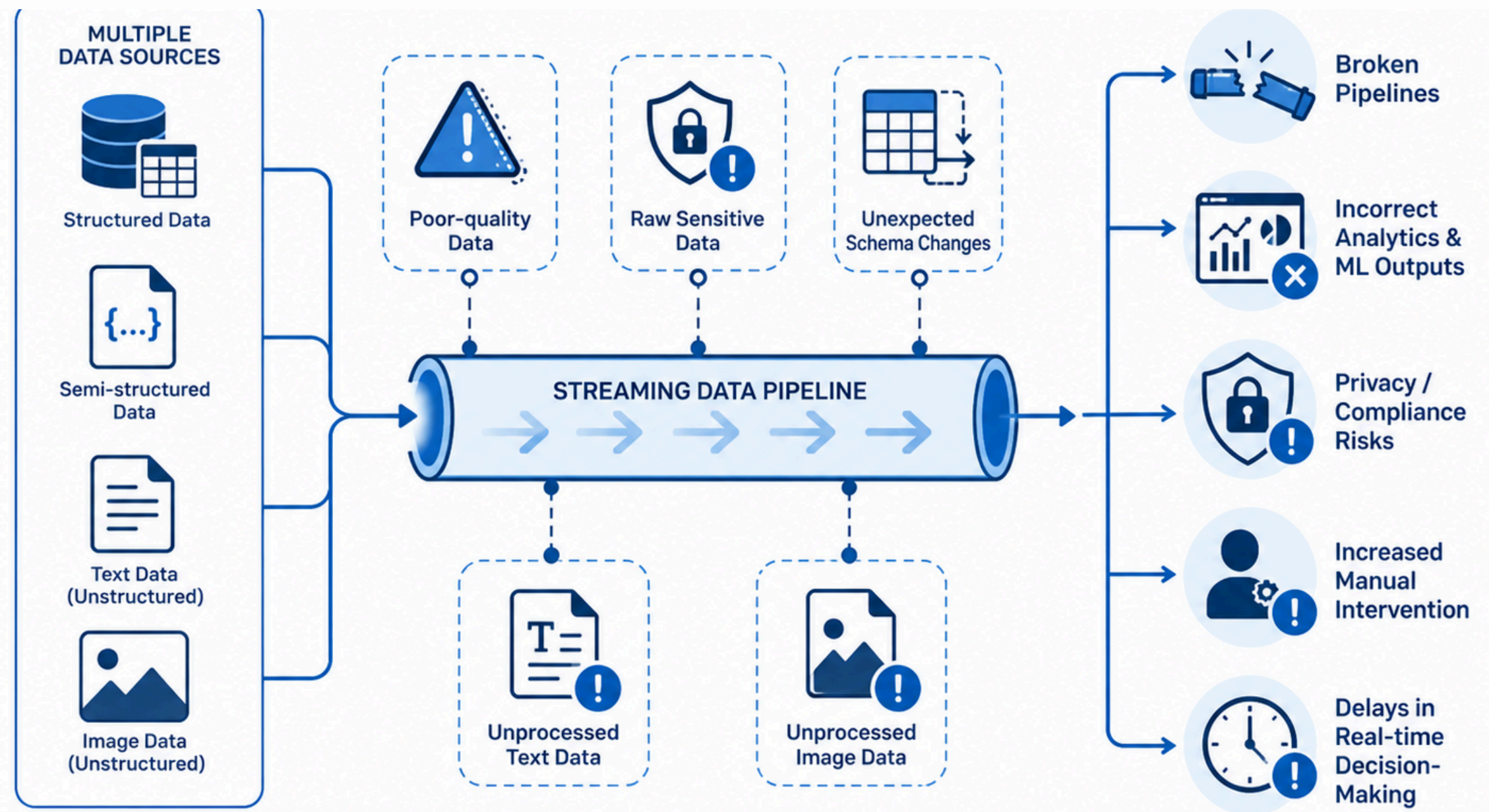


Nanduni



RESEARCH PROBLEM

Modern streaming pipelines continuously receive structured, semi-structured, and unstructured data from multiple sources.



OUR SOLUTION

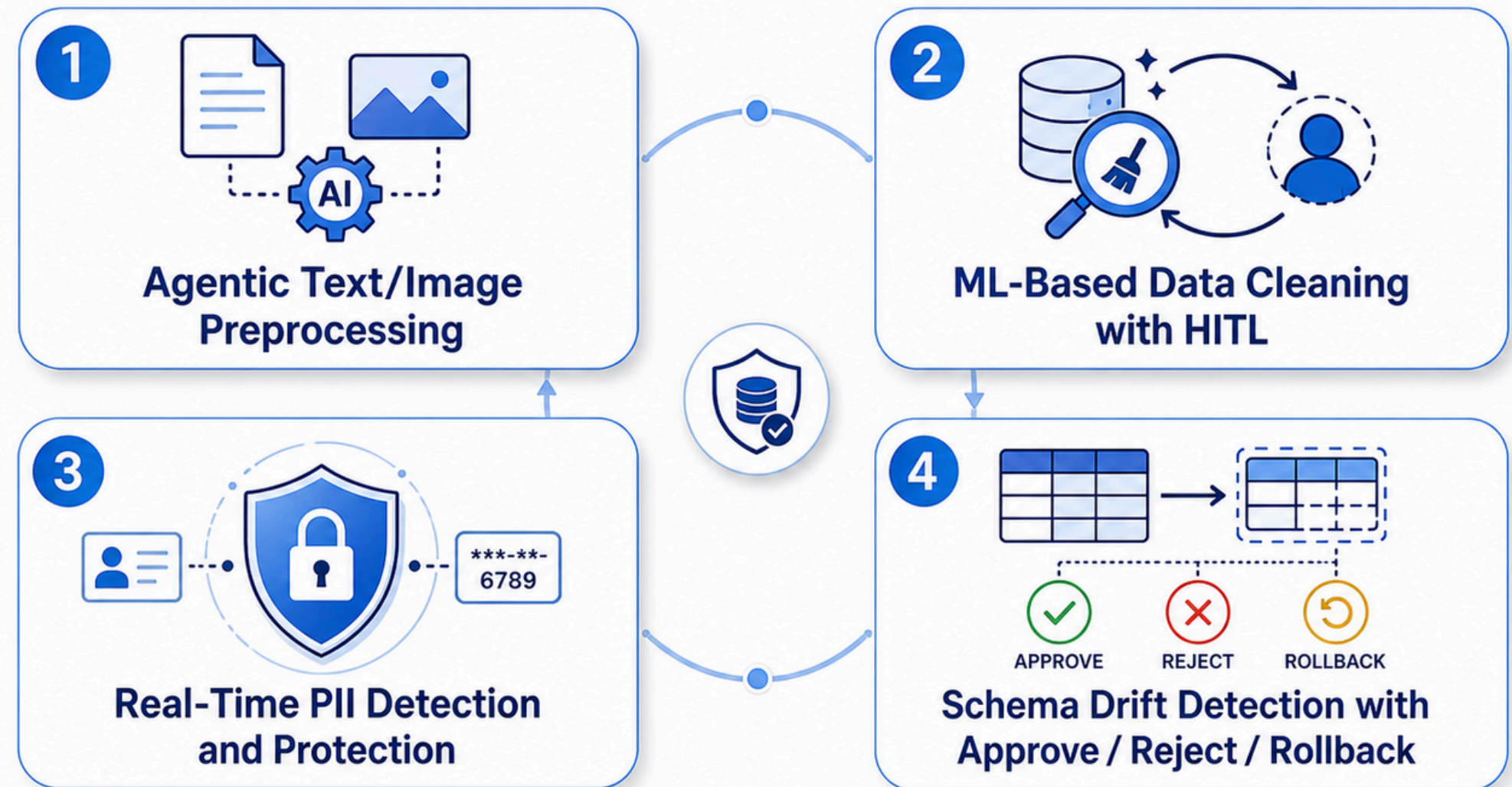
Existing solutions handle quality, privacy, preprocessing, and schema drift separately, mostly after data is stored.

Our solution introduces a modular 4-component framework that applies governance at the ingestion layer, before unreliable or sensitive data reaches storage.



4-Component Framework

Governance at the ingestion layer





IT22549136 | A. S. SILVA

Agentic Multi-Modal Unstructured Data Preprocessing Framework



AGENTIC MULTI-MODAL UNSTRUCTURED DATA PREPROCESSING FRAMEWORK

- Uses an LLM-guided planner to select required text preprocessing agents
- Handles HTML stripping, language detection, translation, spell correction, tokenization, NER, and validation
- Applies enhancement, denoising, deblurring, and ROI extraction only when needed
- Reduces unnecessary processing and improves real time efficiency





IT22893666 | L.T.E. ARACHCHI

Structured Data Cleaning Framework



STRUCTURED DATA PREPROCESSING FRAMEWORK

- Detects structured data issues in real-time streaming batches.
- Inter-column dependency analysis.
- Per-column Random Forest, SVR, and Linear Regression models for context-aware corrections.
- SHAP & LIME explainability for every suggested fix.
- Human-in-the-Loop interface to approve, reject, or override corrections.





IT22556684 | H. A. A. K. ARANGALLA

Automated Privacy Preservation in Data Pipelines



AUTOMATED PRIVACY PRESERVATION IN DATA PIPELINES

- Detects PII using NER + regex
- Covers names, NIC, phone, bank account no.s, credit card no.s, password, location and card data
- GDPR, HIPAA, PDPA policy aware
- Applies masking, hashing, encryption, tokenization, and bcrypt
- Updates privacy rules using RAG-based policy enhancement





IT22204752 | W.R.W.M.N.S. WEERAKOON

Automated Real-Time Schema Drift Detection



AUTOMATED REAL-TIME SCHEMA DRIFT DETECTION

- Detects added, removed, renamed, type-changed, and nullable columns.
- Uses column normalization to reduce false drift alerts.
- Applies multi-feature rename detection.
- Calculates risk using dataset and field-level scoring.
- Sends risky data to staging for human review.
- Supports approve, reject, and rollback with audit logs.



PP2 COMMENTS AND ACTIONS TAKEN

Comment	Action Taken
Why "plugins"? These seem more like libraries or services.	Delivered as 4 independent microservices; can be used individually or sequentially by data engineers.
Spell correction may interfere with data meant to analyze user literacy.	Spell correction output is stored as a separate MongoDB column; original data remains unchanged.
Training one model for all dataset types may not be optimal.	Added a model selection stage — Linear Regression for linear, SVR for smooth non-linear, and Random Forest for noisy/highly non-linear data.

User Acceptance



Research Paper Acceptance

- 18th INTERNATIONAL CONFERENCE on Electronics, Computers and Artificial Intelligence (ECAI 2026)
H-index = 21

ECAI-2026 notification for paper 0423

Inbox x

ECAI-2026 <ecai2026@easychair.org>

11:44 (2 hours ago)

☆

😊

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⋮

to me ▾

Attention :
Add only: 979-8-3315-5818-5/26/\$31.00 ©2026 IEEE;

Dear Amanda Arangalla,

We are glad to inform you that your paper „An Agentic Multi-Modal Unstructured Data Preprocessing Framework with LLM-Guided Orchestration and Quality-Driven Validation” has been accepted to be presented at ECAI conference.

- 2026 The 9th International Conference on Artificial Intelligence and Big Data (ICAIBD 2026)
H-index = 17

2026 The 9th International Conference on Artificial Intelligence and Big Data

Chengdu, China | May 29–31, 2026

ICAIBD

Notification of Acceptance

Paper ID:	IB-2212
Author(s):	Amodhya Silva, Lakmini Abeywardhana, Lavindi Endera Arachchi, Nanduni Weerakoon, Amanda Arangalla
Paper Title:	An Agentic Multi-Modal Unstructured Data Preprocessing Framework with LLM-Guided Orchestration and Quality-Driven Validation



Applied Competitions

- SLASSCOM National Ingenuity Awards (SNIA) 2026 i
- Yarl Geek Challenge (YGC) Season 15 Senior Pitch Competition

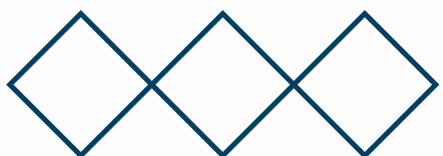
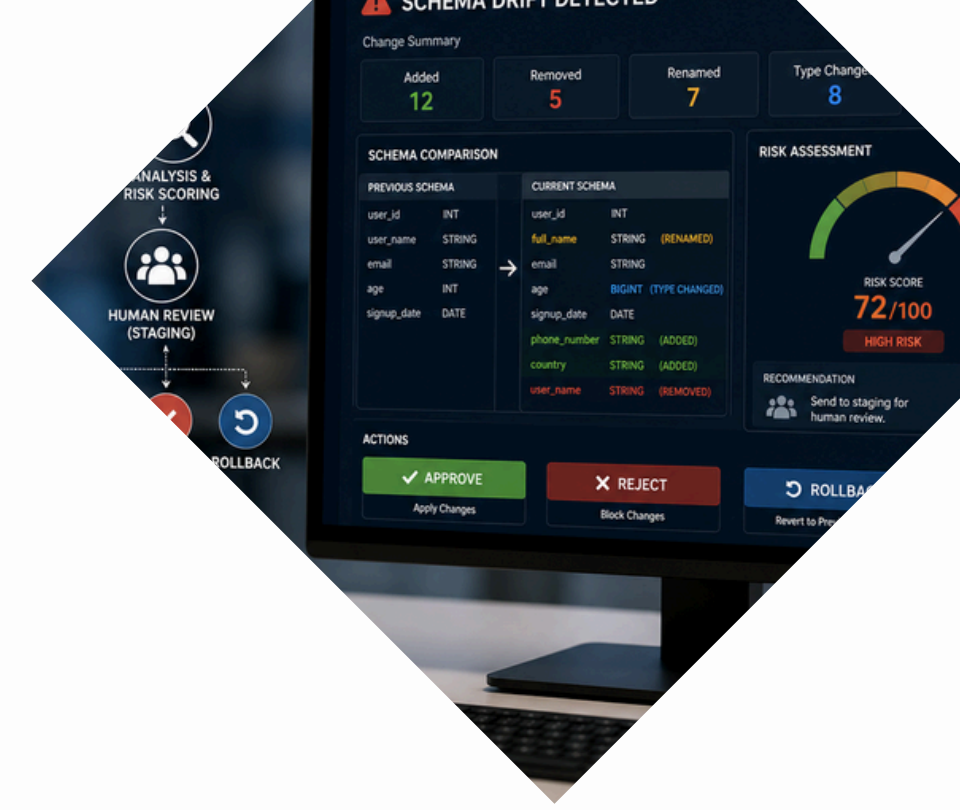


LIMITATIONS

- No automated orchestration between microservices
- Performance may degrade with very high-velocity data streams
- Limited adaptability to highly domain-specific datasets without retraining

FUTURE IMPROVEMENTS

- Implement an orchestration layer to automate the full pipeline.
- Optimize for high-velocity streaming with distributed processing.
- Enable continuous learning to adapt to new data patterns over time.



**THANK
YOU !**

